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Effect of Protic and Aprotic Solvents on the Mechanism of Cellulose Dissolution in Ionic Liquids: A Combined Molecular Dynamics and Experimental Insight

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The present study is undertaken to reveal the effect of co-solvents on the dissolution behaviour of cellulose in 1-ethyl-3-methylimidazolium acetate ([Emim][OAc]) by both quantum chemical (QC) and Molecular Dynamic (MD) simulations. Two categories of cosolvents namely protic and aprotic solvents have been selected in order to enhance the cellulose dissolution in ionic liquid (IL). In the typical IL/cosolvent system, anion of the IL was found to be strongly solvated by the protic solvents (Formamide and Acetamide) thereby resulting in decreasing cellulose solubility, whereas it gave contrary trend for aprotic solvents (DMSO, DMF, DMAc). The dissolution of cellulose was essentially quantified by the cellulose-anion interactions by studying the non-bonded interaction energies. Furthermore, the solid-liquid equilibria experiments were conducted for cellulose in IL and IL/cosolvent mixtures to confirm the simulated results. From MD simulations and experimental results, DMSO was found to be the best cosolvent for the enhancement of cellulose dissolution in IL. The regenerated cellulose was then characterized by Fourier transform spectroscopy (FTIR), X-ray diffraction (XRD) and thermal gravimetric analysis (TGA). From the TGA analysis, thermal stability of regenerated cellulose was found to be reduced by 6 °C at room temperature.

Introduction

Recently, lignocellulosic biomass has been identified as suitable feedstock for the production of fuels and chemicals.[1–3] This is required due to the high utilization of fossil fuels which has increased the production of carbon dioxide.[1,4] Lignocellulosic biomass is constituent of complex composite polymeric carbohydrates (cellulose and hemicellulose) and lignin. Cellulose is the primary component of any lignocellulosic biomass and possess 30–50% of its composition.[2,5] It is also cheap and the most abundant organic matter on the earth.[6] Cellulose has three hydroxyl groups in each glucose molecule and is combined at β(1→4) position.[7] Cellulose molecules have strong inter- and intra-molecular hydrogen linkages through the hydroxyl groups thereby forming crystal structure at a typical condition.[8–10] Cellulose has more affinity to water where it can be dissolved at high temperatures.

Cellulose is hardly soluble in conventional organic solvents because of its numerous intra- and inter-molecular hydrogen bonds.[11,12] Dissolution of natural cellulose is hence a more demanding task under relative mild operating conditions. From an economic point of view, low energy intensive operations are the need of the hour so as to reduce the carbon imprint.[13,14] Solvents such as dimethylacetamide/lithium chloride, N_2O_4-N, $LiClO_4{\cdot}3H_2O$, $LiSCN{\cdot}2H_2O$ and N-methylmorpholine-N-oxide/H_2O have been used for cellulose dissolution.[15–17] However, these solvents have suffered from toxicity, volatility, operation at high temperatures, uncontrollable side reactions, instability, and possess difficulty in solvent recovery.[18] Interestingly, the study by Robin Rogers in 2002 opened a new pathway for cellulose dissolution using Ionic Liquids (IL's). These solvents were found to dissolve cellulose in high amounts at mild operating conditions.[19]

Nowadays, IL's has gained more attention towards biorefining operations due to its attractive properties.[20,21] It exhibits properties such as negligible vapour pressure, non-flammability, and a high thermal and chemical stability.[22,23] IL's can be used as catalyst in many reactions and also used as an alternative to the regular volatile organic solvents.[24,25] The advantages of IL in these processes are that it can easily dissolve biomass and its derived components at lower operating conditions. Further, a slight modification in cation and/or anion structure would result in an enormous range of potential IL's, encompassing a wide variety of solvent properties.[4,26] On the contrary, the disadvantage of IL's are its high viscosity which results difficulties in the processing and handling of dissolution.[5,27]

For solvents, it is a known fact that an increase in viscosity occurs due to the addition of polymers. Hence the mass transfer resistance restricts the amount of polymer which can be dissolved in solvent such as IL.[13] To overcome this phenomenon, protic or aprotic solvents can be added as cosolvent so as to reduce the viscosity of IL's.[28–30] With the less viscosity, higher mass transfer rates can be achieved. The addition of highly po-

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lar cosolvent to IL's has proven to significantly increase the cellulose dissolution at lower conditions.[18] Further when IL's are used as a solvent, there is also a possibility for regeneration and reutilization which reduces the process cost.[6,17]

In this context, the current study reports the dissolution mechanism of cellulose in 1-ethyl-3-methylimidazoilum acetate ([Emim][OAc]) and [Emim][OAc]/cosolvent mixtures with the aim of increasing cellulose solubility in IL's. The protic (formamide (FRM), acetamide (AcM), Ethanol, and Water) and aprotic (dimethylsulfoxide (DMSO), N,N-dimethylformamide (DMF), N,N-dimethylacetamide (DMAc), and acetone) solvents have been chosen to study the role of cosolvents. Initially, Quantum Chemical (QC) calculations are employed to reveal the anion and cation interactions with cosolvents. The dissolution of cellulose in IL and IL/cosolvent mixtures are then determined by following two approaches. The first approach was to further apply molecular dynamic (MD) simulations for optimizing the potential cosolvent with IL through structural properties. The second approach is to confirm the MD simulations results by solid-liquid equilibria (SLE) experiments within a temperature range of 27–90 °C at one hour reaction time.

Results and Discussion

Quantum chemical calculations for the solvation of ions by different cosolvents

The chemical structures of cellulose, IL and cosolvents with atom notations are presented in the Figure 1. The experimental section and computational details of quantum chemical calculations and molecular dynamic simulation were given in the supporting information. Optimized structures of the lowest energy conformers for [Emim]$^+$ and [OAc]$^-$ with different cosolvents are presented in Figure 2 ([OAc]$^-$), and Figures S1-S3 ([Emim]$^+$). Figure 2 shows the lowest energy conformer for [OAc]$^-$ with different cosolvents. It is clearly evident that the O atom of [OAc]$^-$ forms weak hydrogen bonds (C–H…O) with aprotic solvents while the contrary is seen with protic solvents. It should be noted that a weak hydrogen bond is denoted by a bond distance greater than 2 Å.[44] Hence in these systems, the length of the C–H…O hydrogen bond was found to be ~2.15-2.83 Å. On the other hand, the hydrogen bond length of the N–H…O (FRM and AcM) and O–H…O (water and ethanol) is ~1.84 Å and ~1.79-2.08 Å, which indeed indicates strong hydrogen bonding between [OAc]$^-$-protic solvents.

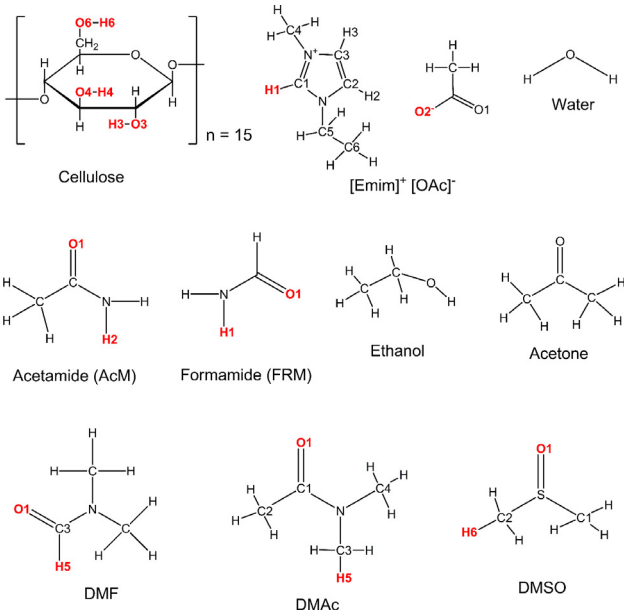

Figure 1. Chemical structures and atom notations of the repeat units of glucose in cellulose (DP = 15), ionic liquid, and different cosolvents used in this work.

In addition, the optimized geometries of [Emim]$^+$-cosolvent pairs are presented in Figures S1-S3. It should be noted that the protons at C1, C2, and C3 sites of the [Emim]$^+$ cation could potentially form hydrogen bonds with O atom of cosolvents. These will be inherently weaker than O–H…O and N–H…O. The formation of a hydrogen bond is due to the transfer of charge between the molecules. This can be affirmed by the computation of NPA (natural population analysis) charges for the isolated cosolvent molecules and ion-cosolvent pairs. The aggregate NPA charge for the cosolvents and ion-cosolvent pairs are given in Table S9. It can be seen that there is a diminutive charge transfer for ion-cosolvent pairs. To support this claim, comparison was made for protic and aprotic solvents with [OAc]$^-$ where the protic solvents was found to exhibit a higher charge transfer. The charge transfer was found to decrease AcM > ethanol > FRM > water > DMSO > DMF > DMAc > acetone. Thus from both cation and anion perspective, C–H…O hydrogen bond has a lower charge transfer than N–H…O and O–H…O.

Further confirmation can be visualized from Table 2, where the interaction energies of [OAc]$^-$-cosolvents and [Emim]$^+$-cosolvents are represented for different cosolvents. From Table 2, the interaction energies of [OAc]$^-$ anion with protic solvents were found to be higher than that of [OAc]$^-$-aprotic solvent pairs. On the other hand, the [Emim]$^+$ and [OAc]$^-$ exhibits approximately similar interaction energy with aprotic solvents. This is due to the fact that the anion is more preferentially solvated by protic solvents, while for aprotic solvents the solvation are of similar order for both ions. It should be noted that, lower interaction energy is preferable as it allows a free movement of the anions. This allows the anions to interact with the incoming cellulose molecules.

Optimization and influence of IL to cosolvent molar ratio

Optimization of molar ratio

In the MD simulations, the cosolvents such as DMSO, DMAc, DMF, FRM, and AcM have been considered based on their on their physiochemical properties and relative or lower interaction with ions. So in the process we have eliminated three

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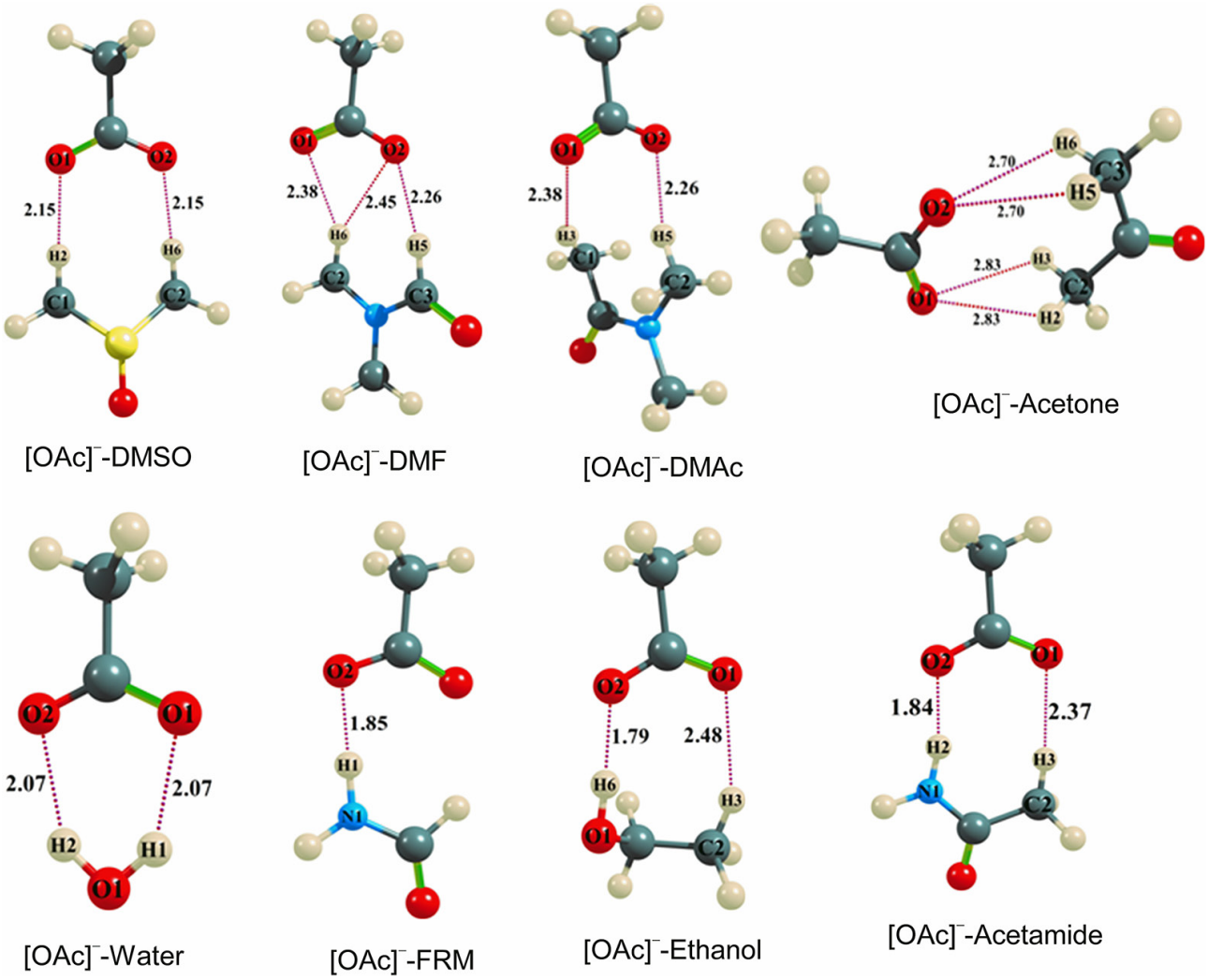

Figure 2. Optimized geometries of ion – molecule pairs of anion [OAc]⁻ with different cosolvent molecules. The dashed lines indicate hydrogen bonds in these pairs. C (cyan), H (white), O (red), S (yellow), and N (blue) are colour representation for different atoms.

Table 1. Non-bonded Interaction Energies for Cellulose-[Emim][OAc]/Cosolvents Obtained from MD Simulations in Different Systems

Simulated System	Ionic Liquids	Cosolvent	R[a]	E_{elec}[b]	E_{vdw}[c]	E_{inter}[d]
Cellulose-IL/DMSO	500	500	1.0	-75290.27	-9822.70	-85112.97
	400	800	2.0	-79872.20	-11491.03	-91363.24
	360	900	2.5	-79543.04	-11033.49	-90576.48
	320	960	3.0	-78183.59	-11857.20	-90040.80
	250	1000	4.0	-76060.164	-12588.60	-88620.65
	225	1125	5.0	-76021.94	-11809.33	-87831.27
Cellulose-IL	300	-	-	-30389.76	-3306.01	-33695.78
Cellulose-IL/DMAc	400	800	2.0	-52532.31	-6905.59	-59437.908
Cellulose-IL/DMF	400	800	2.0	-67249.80	-9465.82	-76715.62
Cellulose-IL/FRM	400	800	2.0	-48929.43	-2395.11	-51324.546
Cellulose-IL/AcM	400	800	2.0	-33347.82	-4693.91	-38041.737

[a] Molar ratio of cosolvent to ionic liquid; [b] electrostatic interaction energy of the system in kcal mol^{-1}; [c] vander Waals interaction energy of the system in kcal mol^{-1}; [d] sum of E_{elec} and E_{vdw} in kcal mol^{-1}

cosolvents namely acetone, water and ethanol. For a benchmark case DMSO was chosen to optimize the molar ratio (R_{DMSO}). In such a study, the molar ratio of [Emim][OAc] to DMSO was varied from 1–5. To recommend the optimum cosolvent molar ratio, we have computed the non-bonded energy between anion, cation, and DMSO with cellulose. The MD simulated interaction energies (sum of electrostatic and van der Waals energies) are given in Table 1. It is seen that the electrostatic interactions governs the non-bonded interactions. From Table 1, it can be observed that the highest interaction energy for [Emim][OAc]/DMSO mixture with cellulose is obtained at $R_{DMSO} = 2$. So here we accomplish the second stage namely to fix the IL/cosolvent

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Table 2. Interaction Energies (kJ mol^{-1}) of [OAc]$^-$ Anion and [Emim]$^+$ Cation with Different Cosolvents Obtained from Quantum Chemical (QC) Calculations.

Entry	Cosolvent	[OAc]$^-$ I.E.	[Emim]$^+$ I.E. (I)	I.E. (II)	I.E. (III)	I.E. (IV)
1.	Ethanol	-74.26	-47.15	-36.95	-38.91	-33.14
2.	Water	-80.30	-46.48	-37.15	-36.65	-45.85
3.	Formamide (FRM)	-101.67	-67.32	-55.40	-56.61	-66.99
4.	Acetamide (AcM)	-99.66	-71.13	-58.58	-60.21	-70.92
5.	DMSO	-77.50	-83.40	-69.17	-70.30	-80.96
6.	DMF	-59.62	-73.01	-58.99	-60.21	-52.18
7.	DMAc	-58.45	-78.90	-61.92	-60.1	-73.47
8.	Acetone	-43.10	-59.67	-58.80	-44.73	-58.85

molar ratio by MD simulations. It should also be noted that, when DMSO is added to the complex system, the change in the electrostatic energy is much higher than the van der Waals energy which implies it is indeed the electrostatic charges which decides the dissolution. It was also observed that an increase in [Emim][OAc] to DMSO molar ratio (i.e., from 2.5 to 5) results in decrease of interaction energy. This affirms the MD derived optimized molar ratio namely 2.0. In the preceding sections, we further proceed to establish the mechanism for the interaction of the ions(of Ionic Liquids) with Cellulose and DMSO.

Interaction of ions with cellulose

The dissolution process is seen to follow two opposite pathways. While on one side, the cation decreases the cellulose solubility due to its strong binding with anion, on the other hand, anion helps in providing hydrogen bonds with incoming hydroxyl groups of cellulose.[18,31,32] In the current scenario our goal is to enhance the cellulose-anion interactions by decreasing the anion-cation interactions using cosolvents. Figure 3 shows the interaction energy of anion, cation, and DMSO with cellulose at different IL to DMSO molar ratios. From Figure 3, it is clear that at $R_{DMSO}=2$, the interaction energy between cellulose and [OAc]$^-$ (-2791.55 kJ mol^{-1}) is higher than the other molar ratios in cellulose-[Emim][OAc]/DMSO systems. The interaction energy of cellulose-[Emim]$^+$ was also found to decrease with increasing cellulose-[Emim][OAc]/DMSO molar ratio. On the contrary, the cellulose-DMSO interaction energy is found to increase with molar ratio. It was observedthat the interaction energy of cellulose with [OAc]$^-$ increases from $R_{DMSO}=1$ to $R_{DMSO}=2$, then decrease from $R_{DMSO}=2.5$ to $R_{DMSO}=5$ due to the higher interaction energy (Figure 4).

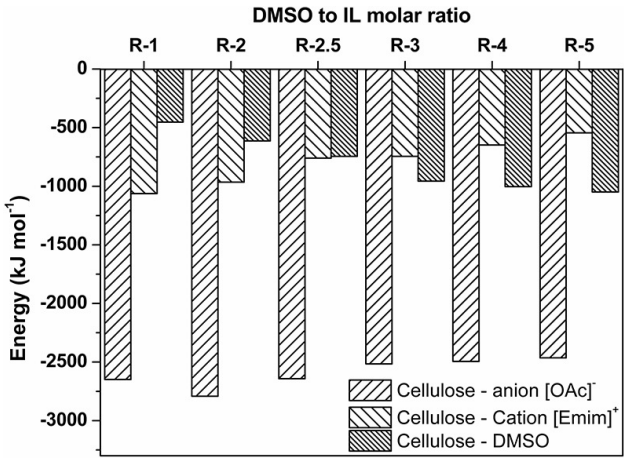

Figure 3. Interaction energies of cellulose with anion and cation obtained from MD simulations at different molar ratio's of IL to DMSO in the cellulose-[Emim][OAc]/DMSO system.

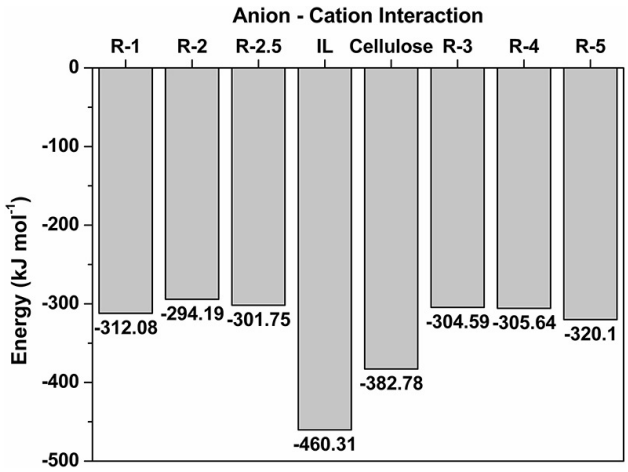

Figure 4. Interaction energies between anion and cation in [Emim][OAc], cellulose-[Emim][OAc] and cellulose-[Emim][OAc]/DMSO systems obtained from MD simulations.

Interaction of DMSO and ions

The non-bonded interaction energies between [OAc]$^-$ and [Emim]$^+$ in [Emim][OAc], cellulose-[Emim][OAc], and cellulose-[Emim][OAc]/DMSO systems are presented in Figure 4. The interaction energy of [OAc]$^-$ with [Emim]$^+$ was found to be lower in cellulose-[Emim][OAc]/DMSO (-294.19 kJ mol^{-1}) mixture than neat [Emim][OAc] (-460.31 kJ mol^{-1}) and cellulose-[Emim][OAc] pair (-382.78 kJ mol^{-1}) (Figure 4). This follows the fact that, strong ionic association is usually present in neat IL due to its inherent electrostatic and hydrogen bonding energies.[33] To dissociate this strong ionic interactions, cosolvent is added to the [Emim][OAc]. The reduction of interaction energy in the presence of DMSO will aid in more production of free ions. As a result, it is the production of free ions which is responsible for

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the formation of strong hydrogen bonds with cellulose. However, the interaction energies of $[OAc]^-$ with DMSO was found to be less than that of $[Emim]^+$-DMSO energies at lower molar ratios (i.e. $R_{DMSO}=1$, 2, 2.5, and 3) (Figure S4). Further an increase in the IL to DMSO molar ratio (i.e. $R_{DMSO}=4$ and 5), resulted in higher interaction energies of $[OAc]^-$-DMSO (Figure S4). This outcome proposes the fact that the degree of solvation of anion is higher than that of cation at higher molar proportions which leads to decrease in cellulose solubility. From this study, it is clearly obvious that the recommended molar ratio for IL to cosolvent is 2. Hence, MD simulations were performed for mixtures of cellulose with [Emim][OAc]/different cosolvents (protic and aprotic) at $R=2$. We now proceed with the studies for other cosolvents.

Effect of protic and aprotic cosolvents

Cellulose-Anion interactions in cellulose-[Emim][OAc]/cosolvent

After the successful optimization of [Emim][OAc] to DMSO molar ratio, a series of MD simulations were carried for the mixture of cellulose-[Emim][OAc] with different cosolvent systems. The non-bonded interaction energies of $[OAc]^-$, $[Emim]^+$ and different cosolvents with cellulose were computed and compared with the cellulose-[Emim][OAc] system. To predict the protic and aprotic solvents effect on cellulose solubility, the interaction energies of cellulose with $[OAc]^-$ were investigated and presented in Table 3. From Table 3, the electrostatic interactions

Table 3. Non-bonded Interaction Energies (kJ mol^{-1}) Between Cellulose and $[OAc]^{<M->}$ Obtained from MD Simulations in Different Cosolvent Systems[a]

Simulated System	E_{elec}[b]	E_{vdw}[c]	E_{Inter}[d]
Cellulose-[Emim][OAc]/DMSO	-3029.13	237.59	-2791.54
Cellulose-[Emim][OAc]/DMAc	-2820.04	323.65	-2496.40
Cellulose-[Emim][OAc]/DMF	-2805.06	297.64	-2507.42
Cellulose-[Emim][OAc]	-2628.06	142.45	-2485.61
Cellulose-[Emim][OAc]/FRM	-2436.83	187.20	-2249.63
Cellulose-[Emim][OAc]/AcM	-2337.91	194.34	-2143.58

[a] The non-bonded interaction energies are calculated based on the per mole of cellulose, [b] electrostatic interaction energy, E_{elec}; [c] vander Waals interaction energy, E_{vdw}; [d] sum of E_{elec} and E_{vdw}

were found to be higher than the van der Waals interactions, signifying that the electrostatic interaction is the controlling parameter for cellulose-$[OAc]^-$ interactions. It was also observed that the interaction between cellulose-$[OAc]^-$ increases with aprotic solvents such as DMSO, DMF, DMAc. This depicts an opposite trend with the addition of protic solvents (FRM, AcM). For either solvents, the interaction energy between cellulose and $[Emim]^+$ was found to decrease with the addition of cosolvent in cellulose-[Emim][OAc] (Figure 5). However the aprotic cosolvents shows a moderate effect on the cellulose interactions in cellulose-[Emim][OAc]/cosolvent system (Figure 5).

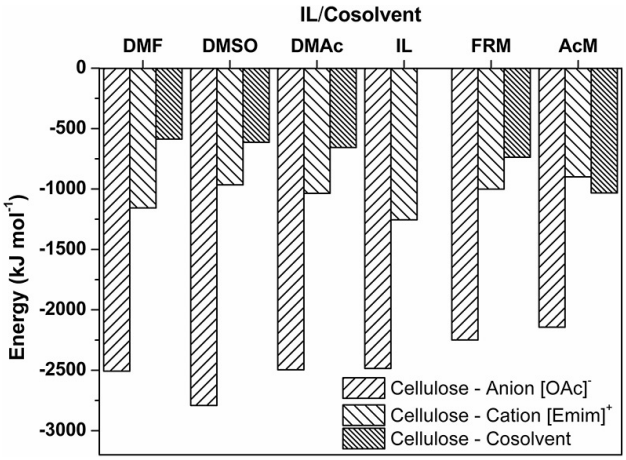

Figure 5. Interaction energies of cellulose with anion, cation and cosolvents in the cellulose-[Emim][OAc]/cosolvents systems obtained from MD simulations.

Anion-Cation interaction in cellulose-[Emim][OAc]/cosolvent

To explore the cosolvent effect on ionic dissociation of IL (i.e. cation and anion), the interaction energies between $[OAc]^-$ and $[Emim]^+$ in cellulose-[Emim][OAc] and cellulose-[Emim][OAc]/cosolvent system was calculated and presented in Figure 6. In

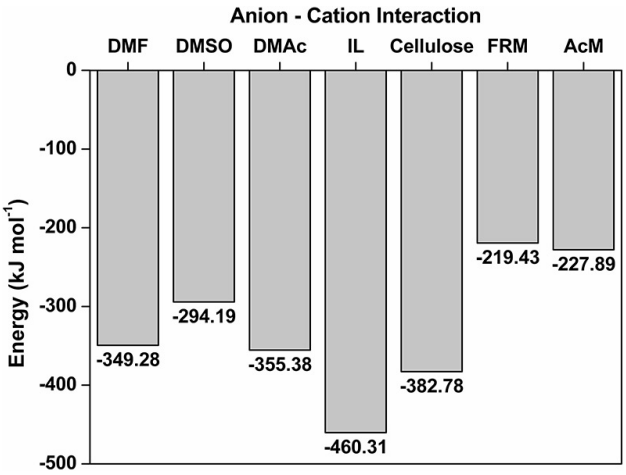

Figure 6. Interaction energies between anion and cation in pure [Emim][OAc], cellulose-[Emim][OAc] and cellulose-[Emim][OAc]/cosolvents obtained from MD simulations.

the cellulose-[Emim][OAc] mixture, the interaction energy between $[OAc]^-$ and $[Emim]^+$ was relatively higher than cellulose-[Emim][OAc]/cosolvent systems (see Figure 6). This suggests a higher association for the cation-anion pair in the cellulose-[Emim][OAc] mixtures. Hence, when a cosolvent is added to the cellulose-[Emim][OAc] mixture, the association or binding energy between anion and cation is partially broken thereby releasing more free ions. The protic solvents (FRM and AcM) were found to dissociate $[OAc]^-$ and $[Emim]^+$ to a larger extent than

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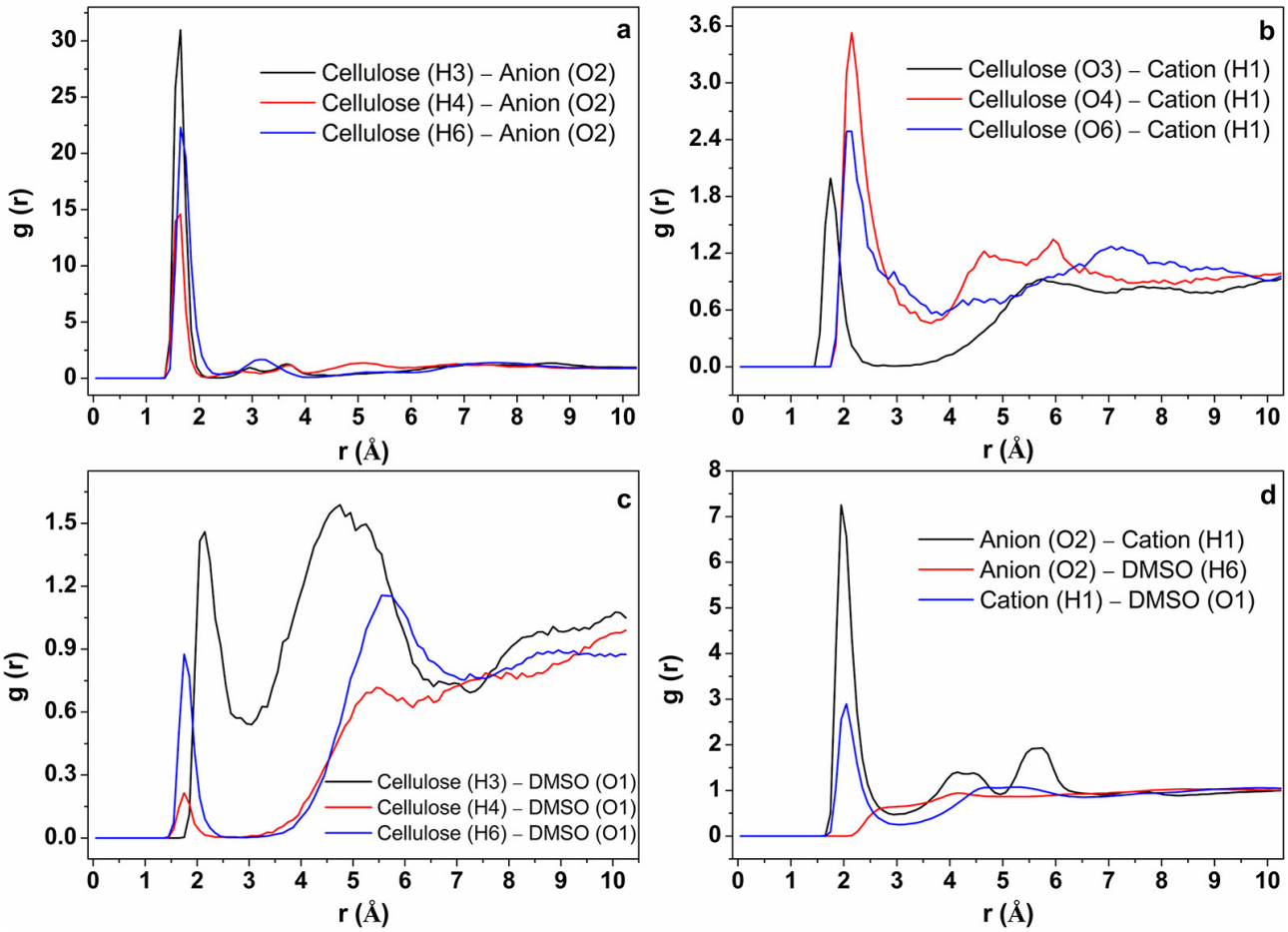

Figure 7. RDF between cellulose molecule with (a) anion (O2), (b) cation (H1), and (c) DMSO (O1) around the selected hydroxyl groups (O3, O4, and O6; see Figure 1) of cellulose and (d) RDF between [Emim][OAc]/DMSO in the cellulose-[Emim][OAc]/DMSO system.

the aprotic solvents, nevertheless they exhibited lower interaction energy between cellulose and $[OAc]^-$ (Table S10). Due to this reason, protic solvents are known to decrease the cellulose solubility, whereas, in the case of aprotic solvents, cation of the IL is found to be more solvated (Table S10) and hence increases the cellulose solubility.

The degree of cosolvent effect on ionic energy dissociation was found to decrease in the following order: FRM > AcM > DMSO > DMF > DMAc (Figure 6). The decrease in the anion-cation interactions can be attributed to the molar volume of the cosolvents (FRM (39.88), AcM (55.7), DMSO (74.84), DMF (76.42), and DMAc (93.06)).[34–36] Thus, a lower molar volume of the cosolvent causes a higher decrease in the anion-cation interactions as they can solvate cellulose molecules more efficiently within the given volume by executing more hydrogen bonds. This is due to the fact that the charge per unit surface area or the charge concentration will be higher for solvents with lower molar volume. This implies that the addition of a smaller cosolvent shows an encouraging trend for the dissociation of ionic pairs. We now move ahead and depict the pair-wise relation between different moieties using RDF and coordination number.

Structural properties of cellulose in [Emim][OAc] and [Emim][OAc]/cosolvent mixtures

Radial distribution functions

The RDF plots between the hydroxyl protons of cellulose molecule (H3, H4 and H6 atoms) with O2 of anion and O1 of cosolvent (DMSO) are depicted in Figure 7. Figure 7 also depicts the RDF of hydroxyl oxygen of cellulose with H1 atom of cation. For DMF, DMAc, FRM, and AcM cosolvents, the RDF plots are given in the supplementary information Figures S5-S9. A sharp peak at 1.65 Å in all *g*(r)'s indicates a strong hydrogen bond between the protons of cellulose and O2 atom of anion for both cellulose-[Emim][OAc] and cellulose-[Emim][OAc]/cosolvent systems. In the case of cellulose-cosolvent, the peak was located within 1.95-2.15 Å. This signifies a formation of a weaker hydrogen bond as compared to cellulose-anion (1.65 Å). For cellulose-cation RDF, the peak was found to lie at 2–2.15 Å representing a weak hydrogen bond.

From the RDF plot of anion (O2) with cation (H1), the *g*(r) peak was found at 1.65 Å in pure IL. A similar peak between anion (O2) with cation (H1) in cellulose-IL system was obtained at 1.75 Å (Figure 8). This suggests that the interaction between

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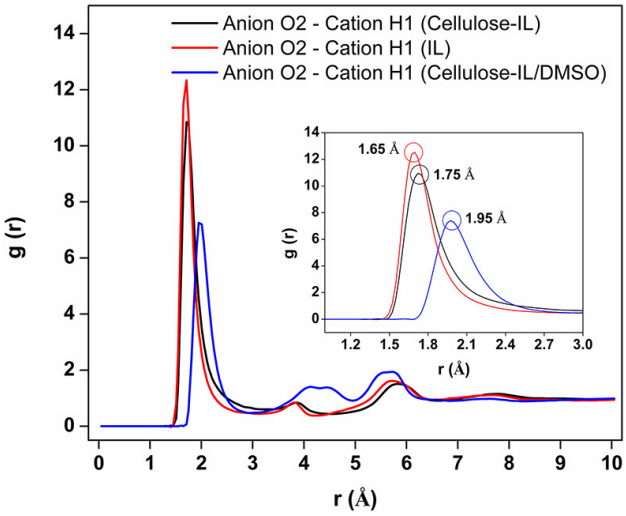

Figure 8. RDF between anion (O2) and cation (H1) of the [Emim][OAc] in the [Emim][OAc], cellulose-[Emim][OAc] and cellulose-[Emim][OAc]/DMSO systems. See Figure 1 for atom notations.

anion and cation has preferably decreased as compared to pure IL interactions. When a cosolvent is added to the cellulose-IL system, the *g*(r) peak has now extended upto 1.95 Å for aprotic solvents and 2.05 Å for protic solvents, respectively (Figure 8 and Figures S5-S9). This extended peak indicate a weak hydrogen bond between $[OAc]^-$ and $[Emim]^+$. This trend is supported by their interaction energies (Figure 6) and coordination number (Table 4) as calculated from MD simulations. We will now take a step ahead and compute the coordination numbers for cellulose-anion and cellulose -IL/cosolvents.

Table 4. Coordination Numbers of First Solvation Shell for Different Systems Obtained from MD Simulations [a]

Simulated System	Cellulose-Anion	Cellulose-Cation	Anion-Cation	Anion-Cosolvent	Cation-Cosolvent
Cellulose-[Emim][OAc]/DMSO	3.84	0.95	1.61	0.52	0.87
Cellulose-[Emim][OAc]/DMF	3.15	1.04	1.94	0.32	0.65
Cellulose-[Emim][OAc]/DMAc	2.91	0.99	2.04	0.28	0.56
Cellulose-[Emim][OAc]	2.87	1.46	4.48	-	-
Cellulose-[Emim][OAc]/FRM	2.73	0.98	0.82	2.40	1.02
Cellulose-[Emim][OAc]/AcM	2.36	0.76	0.91	2.06	1.34
[Emim][OAc]	-	-	5.54	-	-

[a] the coordination number is calculated for a single glucose molecule of cellulose (DP = 15)

Coordination number

The $[OAc]^-$ anion here is found to be surrounded by five to six cations in the first solvation shell of pure IL. This reduces to four cations in the cellulose-[Emim][OAc] system. The respective coordination number was thus found to reduce from 2.04 to 0.91 (Table 4) with the addition of cosolvents. From Table 4, it can be seen that each glucose molecule of cellulose is surrounded by three to four anions in the first coordination shell of cellulose-[Emim][OAc]/DMSO system .In a similar manner, three anions are surrounded in the cellulose-[Emim][OAc]. A comparsion with cellulose-[Emim][OAc]/DMF, DMAc, FRM and AcM cosolvent systems, cellulose gave coordination numbers of 3.15, 2.91, 2.73 and 2.36 anions, respectively in their first coordination shell. For cellulose-cation system, the glucose molecule of cellulose was surrounded by ~1.46 cations in cellulose-[Emim][OAc], while, on account of cellulose-[Emim][OAc]/cosolvent system, the coordination number was found to show a decreasing trend (Table 4). This is due to the fact that the interaction between cellulose and cation is lesser as compared to cosolvent system (Figure 5).

When a cosolvent is added to the cellulose-[Emim][OAc] system, the interaction between $[OAc]^-$ and aprotic solvents decreases, while it increases for protic solvent systems. The RDF peak *g*(r) of $[OAc]^-$ and cosolvents were found to be at 1.75 Å and 2.75 Å for protic and aprotic solvents, respectively (Figures S5-S9 (d)). The RDF peak of $[Emim]^+$ and cosolvents were located at 2.15 Å and 2.05 Å for aprotic and protic solvents, respectively. Thus, it is clear that the protic and aprotic solvents are equally solvated by the $[Emim]^+$ cation. The coordination number for $[OAc]^-$-aprotic solvents were also found to be lower than $[Emim]^+$-aprotic solvents (Table 4). It is thus revealed that the $[Emim]^+$ is more solvated by aprotic solvents than $[OAc]^-$.

A similar effect has been observed by Andanson *et al.*[13] where the effect of cosolvents namely DMSO and water on with 1-butyl-3-methylimidazolium acetate [Bmim][OAc] and cellulose were reported. In their study, they obtained the H-bond formation statistics between hydroxyl groups of glucose units and the different sites of [Bmim][OAc]/cosolvent mixtures.[13] The probability of hydrogen bonds formation between the hydroxyl groups (H3, H4, and H6 as per Figure 1) of glucose and acetate anion were found to be 86.6, 89.2, and 84.2%, respectively in cellulose-[Bmim][OAc]/DMSO mixtures. Similar values for hydrogen bonds involved between glucose and DMSO molecules were between 0.1 to 0.6%. It was reported that the oxygen atom of water molecule tends to accept H-bonds from glucose around 10 times higher than the oxygen atoms in DMSO. These results also imply that the addition of protic solvents decreases the probability of the H-bonds between cellulose and the anion of the ionic liquid which is the opposite for aprotic solvents. In another study[37] with spatial distribution functions, the aprotic solvents were found to be disseminated in a circle along the C1-C2 axis of $[OAc]^-$, while protic solvents are distributed around the O1 and O2 atoms of $[OAc]^-$ anion. Thus similar to our work they concluded that a distribution of protic solvents around the O1 and O2 atom of $[OAc]^-$ hinders

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the interaction between cellulose and anion. Thus in summary the obtained results of the present study are in good agreement with the reported results of Andanson *et al.*[13] and Zhao *et al.*[37].

Experimental dissolution of cellulose in [Emim][OAc] and [Emim][OAc]/cosolvent mixtures

Apart from the evidence of MD simulations, the experiments were also performed for the validation of cellulose dissolution in IL and IL/cosolvent systems. The temperature was kept in a range of 27 °C to 90 °C for 1 h and is shown in Table 5. The dissolution of cellulose was found to increase with temperature. From Table 5, it can be seen that the aprotic solvents tends to dissolve higher amounts of cellulose at ambient temperature. This is higher than neat IL and IL/protic solvent (FRM) systems where $R_{cosolvent}=2$. The addition of protic solvent (FRM) to [Emim][OAc] lowers the cellulose solubility due to the preferential solvation of the anion by FRM (Table S10). As per the observation of Xu *et al.*[18], they found that the addition of water or methanol (protic solvents) to the IL which leads to a decrease in solubility of cellulose. Hence, it is suggested that protic solvents like water or methanol may be recommended for the regeneration of cellulose in IL's.

Table 5. Experimental Solubility of Cellulose in [Emim][OAc] and [Emim][OAc]/Cosolvent System at Selected Molar Ratios with Function of Temperature and Time

Entry	Ionic Liquid/Cosolvent	Condition	Solubility (wt %)
1.	[Emim][OAc]	27 °C, 1 h	2.10
		50 °C, 1 h	6.90
		70 °C, 1 h	9.46
		90 °C, 1 h	14.63
		110 °C, 1 h	19.10
		130 °C, 1 h	27.69
2.	DMSO	27 °C, 1 h	1.22
3.	DMF	27 °C, 1 h	1.15
4.	DMAc	27 °C, 1 h	1.05
5.	FRM	27 °C, 1 h	<1
6.	[Emim][OAc]-DMSO (R=1)	27 °C, 1 h	8.21
7.	[Emim][OAc]-DMSO (R=2)	27 °C, 1 h	9.66
		50 °C, 1 h	14.80
		70 °C, 1 h	17.31
		90 °C, 1 h	22.38
8.	[Emim][OAc]-DMSO (R=2.5)	27 °C, 1 h	7.83
9.	[Emim][OAc]-DMSO (R=3)	27 °C, 1 h	6.35
10.	[Emim][OAc]-DMSO (R=4)	27 °C, 1 h	5.92
11.	[Emim][OAc]-DMSO (R=5)	27 °C, 1 h	4.89
12.	[Emim][OAc]-DMF (R=2)	27 °C, 1 h	7.15
13.	[Emim][OAc]-DMAc (R=2)	27 °C, 1 h	6.30
14.	[Emim][OAc]-FRM (R=2)	27 °C, 1 h	1.41

The experimental solubility of cellulose in [Emim][OAc] and [Emim][OAc]/cosolvent ($R_{cosolvent}=2$) are then compared with MD simulated interaction energies of cellulose with $[OAc]^-$ (Figure 9). The experimental solubility is bound to be affected by many factors such as purity of IL, source of chemicals, cellulose degree of polymerization, moisture content, and temperature of the system. Despite these factors, the simulated energies are in good agreement with the experimental solubility (Figure 9). From Figure 9, it is worthy to note than an inverse relation is obtained between non-bonded interaction energies and the experimental solubilities. This suggests that the adopted MD simulated model has correctly described the intermolecular interactions in the mixtures of cellulose, IL, and cosolvents. Consequently, the simulated interaction energies are helpful in deciding a cosolvent so as to enhance the dissolving capability of IL's for cellulose.

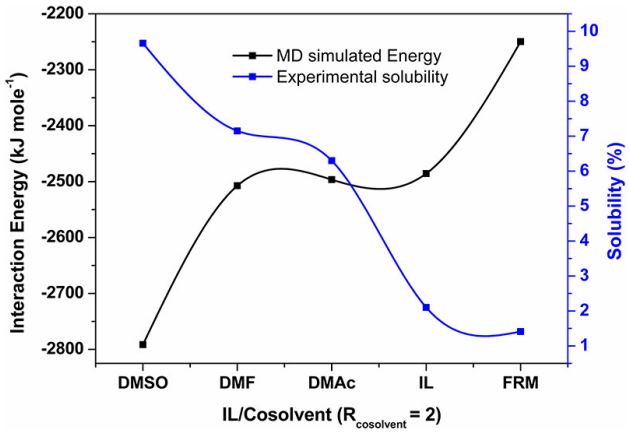

Figure 9. Correlation of experimental solubility and simulated interaction energies between $[OAc]^-$ and cellulose in the cellulose-IL/Cosolvent (R=2) system at room temperature.

The cellulose dissolution order (Figure 9) for cosolvents is thus found to follow: DMSO > DMF > DMAc > FRM. Based on QC, MD and Experiments, we have proposed a dissolution mechanism for the cellulose in [Emim][OAc]/cosolvent systems (Figure 10). The reaction mechanism is described by two steps. In the first step, we assume a strong association pair of ions in pure IL's. Thus, when cosolvents are added to the IL, the strong association is partially destroyed by releasing higher free cations and anions in the solution. In the next and final step, the dissociated free anions interact with cellulose thereby enhancing the cellulose solubility. Hence, from QC, MD and Experiments, it is recommended to use aprotic solvents as it decreases the solvation of anions with its corresponding cation.

The regenerated cellulose was also characterized by XRD, FTIR, and TGA analysis to analyse the chemical structure and thermal decomposition behaviour. The XRD analysis of native and regenerated cellulose is given in Figure 11. The original cellulose is indicated as cellulose I at a diffraction angle (2θ) of 15.1° and 22.6°.[38,39] The regenerated cellulose exhibits similar diffraction patterns except when dissolved in pure IL and IL/aprotic solvents (Figure S13 and S14). The diffraction angle for cellulose in [Emim][OAc]/aprotic cosolvents were revealed at 12.3° and 20.35°. This ascription indicates that the transformations of cellulose I to cellulose II occurrs after the dissolution of cellulose.[40,41] For IL/protic solvent system, the peaks are obtained at 22.65 °C and 15.15 °C which is similar to that of original cellulose. The result of XRD patterns suggests that the crystallinity of cellulose was found to decrease after its dis-

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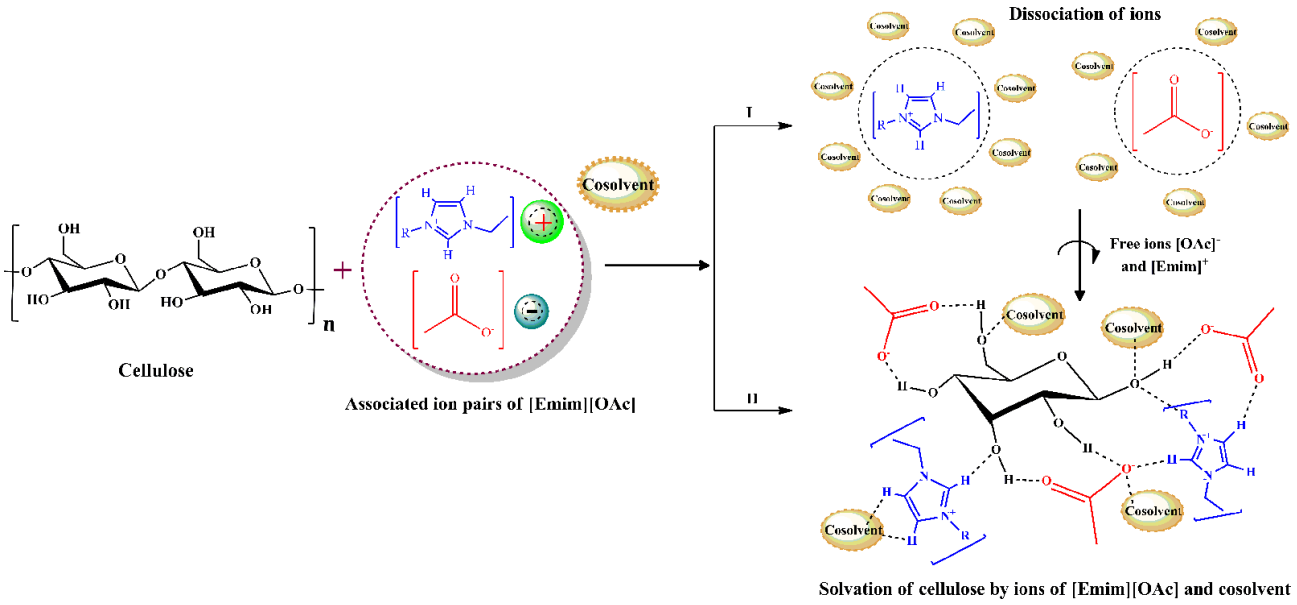

Figure 10. Proposed reaction mechanism for cellulose solvation in IL/cosolvent system, (I) Solvation of ions by cosolvent, (II) Mechanism between cellulose and IL/cosolvent.

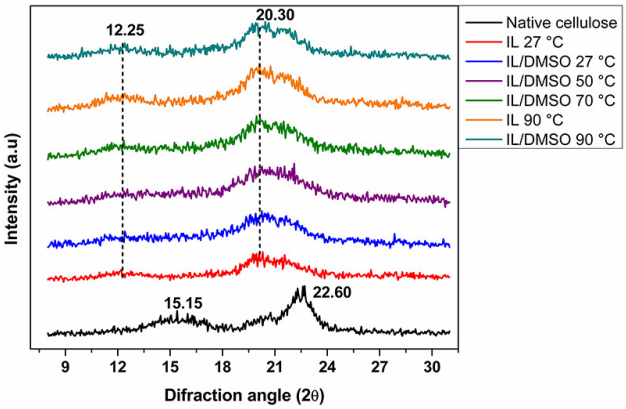

Figure 11. XRD patterns of native cellulose and regenerated cellulose form IL ([Emim][OAc]) and IL/DMSO (R = 2) at different temperatures.

solution with IL and IL/aprotic solvents. On the contrary, IL/protic solvents does not show a similar effect. The XRD patterns of pure IL, cosolvents, and IL/cosolvents ($R = 2$) at different temperatures are presented in the supplementary information (Figure S13 and S14). The FTIR and TGA analysis for the regenerated cellulose are also presented in the supporting information.

Conclusions

The current study reports the effects of IL and IL/cosolvent mixtures for the cellulose dissolution by QC and MD simulations. This is further complemented by experimental studies. Aprotic solvents (DMSO) were found to be good candidates for the dissolution of cellulose in IL's at a molar ratio of two ($R_{DMSO} = 2$). With addition of cosolvent to IL, the strong association pairs of ions are partially destroyed thereby releasing more number of free cations and anions from IL. The dissociated free anions would then interact with cellulose and increase the solubility of cellulose. The interaction energies, RDF plots, and coordination numbers of cellulose-[Emim][OAc]/cosolvent systems also revealed the same mechanism. A lower molar volume of the cosolvent was found to cause a higher decrease in the anion-cation interactions as they can solvate cellulose molecules more efficiently within the given volume by executing more hydrogen bonds. The cellulose dissolution order for cosolvents is thus found to follow: DMSO > DMF > DMAc > FRM. From QC and MD calculations it is recommended to use aprotic solvents as it decreases the solvation of anions with its corresponding cation. However, the protic solvents can be utilized for the recovery of cellulose from IL. The XRD patterns of regenerated cellulose also suggests that the IL/aprotic solvents mixtures does indeed disturb the cellulose crystalline structure. In summary, the solubility of cellulose is mainly determined by the free concentration of anions in IL system.

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Keywords: Cellulose · Cosolvent · Interaction Energy · Ionic Liquid · Molecular Dynamics

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